Open mapping as critical shared infrastructure

OpenStreetMap represents one of the most successful examples of the open source model applied to data rather than code. Founded in 2004, OSM has grown into a database of **10+ billion nodes** maintained by **2.25 million contributors**, creating geographic data that powers everything from humanitarian disaster response to billion-dollar tech products. The emergence of Overture Maps Foundation in 2022—backed by Meta, Microsoft, Amazon, and TomTom—signals not competition but ecosystem maturation: (Wikipedia) major corporations recognizing that collaborative open infrastructure outperforms proprietary alternatives. For developers, OSM demonstrates that the principles of open source—permissionless contribution, copyleft licensing, distributed governance—work just as powerfully for data as for software.

The founding story: frustration with proprietary map data

Steve Coast founded OpenStreetMap in **July 2004** while a student at University College London, motivated by frustration with the UK's Ordnance Survey. Despite being taxpayer-funded, Ordnance Survey data carried expensive licensing fees and usage restrictions. (Wikipedia) Coast saw an opportunity to apply Wikipedia's collaborative model—then just three years old—to geographic data, creating what he called "a free, editable map of the world." (Wikipedia)

The first street was mapped on **December 10, 2004**, when Coast cycled around London's Regent's Park with a GPS receiver. The project grew slowly at first: the **1,000th user** registered on Christmas Day 2005. Early growth required expensive GPS devices, but a turning point came in **December 2006** when Yahoo allowed aerial imagery tracing, removing the hardware barrier. Microsoft's Bing imagery followed in 2010, accelerating contributions dramatically. (Wikipedia)

The philosophical foundation runs deeper than convenience. OSM treats geographic data as a **commons**—contributed by volunteers, belonging to everyone, licensed to prevent enclosure. This parallels free software's treatment of code: the data exists not as corporate property but as shared infrastructure anyone can use, improve, and redistribute under consistent terms.

The ODbL: copyleft licensing designed for data

OSM uses the **Open Database License (ODbL) 1.0**, a copyleft license specifically designed for databases. (Wikipedia) This choice reflects a crucial insight: traditional software licenses like GPL don't translate well to data, and Creative Commons licenses were designed for creative works rather than factual databases.

The ODbL addresses three layers of legal protection that data requires. Copyright protects the database's structure and arrangement. Database rights (primarily in the EU) provide specific legal protections for compiled data. (Opendatacommons) And a contractual layer provides protection in jurisdictions where neither copyright nor database rights apply. This triple-layer approach ensures OSM data remains protected globally.

The license grants broad permissions: you can copy, distribute, use, and adapt the database commercially with no fees. The key obligations are **attribution** (Wikipedia) ("© OpenStreetMap contributors") and **share-alike** for derivative databases. However, the share-alike trigger is carefully scoped through the distinction between "produced works" and "derivative databases."

A **produced work**—rendered map tiles, routing directions, geocoding results, or printed maps—can be licensed under any terms you choose. (OpenStreetMap Foundation) You're free to build proprietary applications on OSM data. A **derivative database**, however, must remain under ODbL. This means if you modify, correct, or extend OSM data itself—adding new features, fixing coordinates, or combining it with other data in ways that create a new database—the result must be shared openly. (OpenStreetMap Foundation)

This distinction makes ODbL remarkably developer-friendly while maintaining the commons. OSM originally used CC-BY-SA, but after years of community debate, switched to ODbL on **September 12, 2012**. (OpenStreetMap) The transition required individual consent from contributors, with **~75% of data** fully relicensable and only **~1%** lost when approximately 380 contributors declined out of 50,000.

How distributed contribution actually works

OSM's contribution model mirrors open source software projects: permissionless editing with quality maintained through community vigilance rather than gatekeeping. Anyone can create an account and edit within minutes—there's no approval process, no review before changes go live. (OpenStreetMap)

The editing ecosystem offers tools for different skill levels. The **iD editor**, embedded on openstreetmap.org, provides a beginner-friendly web interface with a built-in tutorial. Meta's **Rapid** fork integrates AI-suggested features for verification. (OpenStreetMap) Power users gravitate to **JOSM**, a Java desktop editor with extensive plugins—representing just **6.9% of users** but responsible for **59.1% of all edits**, (OpenStreetMap) demonstrating the typical open source pattern of dedicated contributors driving most output. Mobile apps like **StreetComplete** gamify contribution with quest-based interfaces, (Opensource.com) and **HOT Tasking Manager** coordinates organized mapping by dividing areas into grid squares that volunteers claim and complete. (OpenStreetMap)

OSM's data model uses three geometric primitives: **nodes** (points), **ways** (ordered sequences of nodes), and **relations** (ordered lists combining elements for complex features like routes or multipolygons). (OpenStreetMap Wiki) Attributes use free-form key=value tags (OpenStreetMap) (Wikipedia) following a **folksonomy** approach— (Wikipedia) there's no central authority dictating allowed tags. The wiki documents conventions, but mappers can use "any tags you like." (Wikipedia) With **99,000+ distinct keys** in use, this flexibility enables rapid innovation while community documentation creates de facto standards.

Quality control happens post-hoc through community vigilance. **OSMCha** (the OpenStreetMap Changeset Analyzer) filters and flags potentially problematic edits—large deletions, bulk changes from new users, modifications to sensitive features. (Hotosm) **JOSM's validator** catches errors before upload. The **Data Working Group (DWG)** handles escalated issues, with authority to block accounts and redact problematic data. (OpenStreetMap Foundation) Despite permissionless editing, vandalism constitutes only **~0.2% of edits**—lower than Wikipedia—with most fixed within minutes to hours. (Wikipedia)

Corporate contributors have become significant players. Microsoft provides Bing imagery and released US building footprints. Meta develops AI-assisted mapping tools. Apple's **5,000 staff** participate in Missing Maps volunteering. GIGAZINE +2 Amazon Logistics revises maps based on driver GPS traces. Grab contributed **800,000+ km of Southeast Asian roads** based on driver data. Wikipedia These contributions follow OSMF's **Organized Editing Guidelines** requiring documentation, changeset hashtags, and community engagement.

Overture Maps Foundation: enterprise infrastructure on open foundations

The **Overture Maps Foundation** launched in **December 2022** under the Linux Foundation, founded by Amazon Web Services, Meta, Microsoft, and TomTom. Wikipedia Blog do ITGS It now includes **40+ member organizations** Linux Foundation including Esri, Foursquare, and the Humanitarian OpenStreetMap Team. Geo Week News

Overture is explicitly positioned as complementary to OSM rather than competitive. openstreetmap Overturemaps Its executive director Marc Prioleau stated: "Competition assumes scarcity. With no lack of data to be mapped, resources to do so, and financial resources to fund both initiatives, I see no base for competition." Geo Week News Approximately **40% of Overture records come from OpenStreetMap** OpenStreetMap through the Daylight Map Distribution, a curated monthly release with quality checks. OpenStreetMap

What Overture adds is enterprise-scale processing. It **conflates 200+ data sources**—OSM data, Microsoft's ML building footprints (1B+ buildings), Meta's places data, government datasets, Esri Community Maps, and more—into a unified product. Overturemaps The **schema standardization** provides documented, stable attribute formats (currently v1.12.0) with snake_case formatting for SQL compatibility. Quality validation pipelines perform checks beyond OSM's raw data.

The **Global Entity Reference System (GERS)** provides stable unique identifiers for real-world entities, enabling external data to link to Overture base layers— Linux Foundation solving a fundamental interoperability challenge. Overturemaps Fast Company named GERS one of 2025's "Next Big Things in Tech." Linux Foundation

Overture organizes data into six themes. **Buildings** provides **2.3 billion footprints** worldwide. PR Newswire **Places** contains **64 million POIs**. **Transportation** covers **86 million km of roads**. Linux Foundation **Addresses** includes **446 million address points**. **Base** provides infrastructure, land cover, and water features. **Divisions** contains administrative boundaries in 40+ languages. Overturemaps

The licensing model respects OSM's copyleft requirements through dual licensing. **OSM-derived data** (Base, Buildings, Transportation, Divisions themes) carries the **ODbL**, maintaining share-alike requirements. Overturemaps OpenStreetMap Non-OSM-derived data (primarily Places) uses the **CDLA Permissive 2.0**, allowing unrestricted use. Peace Love Freedom Governance follows Linux Foundation structure with a steering committee of representatives from founding members and membership tiers based on financial and engineering contribution levels.

The virtuous cycle between community and commercial mapping

The relationship between OSM and Overture exemplifies how corporate investment can strengthen rather than undermine open commons. Many Overture members—Meta, Microsoft, TomTom—are simultaneously among OSMF's highest-tier corporate sponsors. Openstreetmap Wikipedia The OSM Foundation acknowledged this in

December 2022: "The resources that Overture's founders are investing into the project, and their stated commitment to OpenStreetMap's work and community, show the value they place on good maps." openstreetmap Openstreetmap

Contributions flow both directions. Companies derive enormous value from OSM data, then contribute improvements back through dedicated mapping teams, imagery, tooling, and direct funding. **About 17% of OSM road kilometers** were last edited by corporate teams as of 2020. Wikipedia Lyft's mapping team made **5.5 million edits** over four years. OpenStreetMap Uber actively contributes across 70+ markets. OpenStreetMap These contributions improve the commons that benefits everyone—including competitors.

Overture explicitly encourages members to contribute to OSM directly: "Map edits that are right for OpenStreetMap should go to OpenStreetMap." openstreetmap +2 The data released by Overture remains available under open licenses, with ODbL preserved for OSM-derived content. The OSMF noted this complementarity: "Our community-driven approach to data collection will remain foundational to any global map." openstreetmap Openstreetmap

Different strengths serve different needs. OSM excels at community-driven ground truth, local knowledge (narrow alleyways, neighborhood POIs), humanitarian response, and real-time updates through distributed volunteers. Overturemaps Overture excels at enterprise-scale quality validation, schema standardization, multi-source conflation, and stable identifiers for commercial applications. Each makes the other more valuable.

OSM powers critical infrastructure worldwide

The scale of OSM adoption demonstrates its transformation from hobby project to essential infrastructure. Major tech companies build products on OSM: **Apple Maps** uses OSM data for some regions. Wikipedia **Meta** uses it for Facebook and Instagram maps. OpenStreetMap **Microsoft Bing** integrates OSM. **Pokémon GO** switched to OSM in 2017, with game mechanics influenced by map data. OpenStreetMap **Snapchat's Snap Map**, powered by Mapbox, runs on OSM. OpenStreetMap **Craigslist** switched from Google Maps in 2012. Wikipedia

Government adoption spans continents. **TriMet** (Portland, Oregon) adopted OSM over a decade ago as its official street network, OpenStreetMap US finding it yields better results than local government GIS datasets. Wikipedia The **French Gendarmerie and Police**, **Italian Carabinieri**, and **Police Scotland** all use OSM. Openstreetmap The **Government of Lesotho** is transitioning its national GIS to OSM. OpenStreetMap Transit agencies across Estonia, Norway, Germany, and cities from Krakow to Malaga run OSM-based journey planners. OpenStreetMap

Humanitarian impact has been transformative since the **2010 Haiti earthquake**, when the OSM community created "the most complete digital map of Haiti's roads" within 48 hours—used by the US Marine Corps for search and rescue. Wikipedia The **Humanitarian OpenStreetMap Team (HOT)**, founded that year, LinkedIn has since completed **7,000+ projects in 150+ countries**. Heigit The **Missing Maps** project, launched in 2014 with Médecins Sans Frontières and Red Cross societies, proactively maps vulnerable areas before crises. Humanitarian OpenStreetMap … During the **2015 Nepal earthquake**, 8,000 OSM volunteers contributed maps used by UN agencies and the Nepal government. Opensource.com In **Sierra Leone's Ebola response**, OSM mapping improved patient residence tracking by 27%. Audaciousproject

Transportation platforms depend heavily on OSM. **Lyft** built its mapping infrastructure on OSM, finding road attributes correct over **95% of the time**. Medium **Uber** uses OSM across 70+ markets. GitHub Wikipedia Ride-sharing giant **Grab** not only uses but significantly contributes to OSM in Southeast Asia. Wikipedia Airlines including **Air France** and **Alaska Airlines** display OSM-based in-flight maps. **Deutsche Bahn** trains show OSM on carriage displays. openstreetmap Openstreetmap

Conclusion

OSM's two-decade trajectory offers lessons for anyone building or consuming open infrastructure. The copyleft ODbL licensing ensures data remains a commons while enabling commercial use—striking a balance that has attracted both passionate volunteers and trillion-dollar corporations. The distributed contribution model demonstrates that permissionless editing can produce world-class data when combined with post-hoc validation and community governance. And the emergence of Overture Maps Foundation shows that major corporate investment can complement rather than compete with community efforts when structured around open licensing and transparent governance. Jawg Maps

For open source developers, OSM represents mapping as the model predicts: not as proprietary lock-in but as shared infrastructure that improves through use. The data exists as a commons. The tools are open source. The governance is community-driven. And increasingly, the companies that could afford to build proprietary alternatives instead invest in making the commons stronger—because in the end, that produces better maps for everyone.